

- The conditions that apply to an EHT MU PPDU shall also apply to a UHR MU PPDU, and
- The conditions that apply to an EHT TB PPDU shall also apply to a UHR TB PPDU.

A non-AP UHR STA that is in intra-PPDU power save mode may enter doze state or become unavailable until the end of a PPDU currently being received if all of the following conditions are met:

- the PPDU is a UHR MU PPDU where the RXVECTOR parameter UPLINK_FLAG is 0,
- the RXVECTOR parameter PPDU_TYPE is 1 or 2,
- the RXVECTOR parameter BSS_COLOR_2 is the BSS color of the BSS in which the STA is associated,
- the RXVECTOR parameters STA_ID do not include the identifier of the STA or the broadcast identifier(s) intended for the STA,
- and the BSS Color Disabled subfield is 0 in the most recently received HE Operation element from the AP with which it is associated.

37.18 Non-primary channel access (NPCA)

37.18.1 General

Non-Primary channel access allows STAs within a BSS to switch to an alternate channel during a period of time when OBSS activity is detected on part of the BSS operating channel.

A non-AP STA that supports NPCA operation is called an NPCA non-AP STA. An AP that supports NPCA operation is called an NPCA AP. An NPCA non-AP STA and NPCA AP shall set the NPCA (#12558)Supported field of the UHR MAC Capabilities Information field of the UHR Capabilities element to 1. A non-AP NPCA STA shall not enable the NPCA mode unless it is associated with an NPCA AP that has enabled NPCA operation.

An NPCA AP that has an operating bandwidth less than 80 MHz shall not enable NPCA operation. An AP of a multiple BSSID set that enables NPCA operation shall indicate the same NPCA primary channel, same NPCA minimum duration, same NPCA switching delay and same NPCA switch back delay as all of the other APs of the same multiple BSSID set that have enabled NPCA operation. An AP of a co-hosted BSS that enables NPCA operation shall indicate the same NPCA primary channel, same NPCA minimum duration, same NPCA switching delay and same NPCA switch back delay as all of the other APs of the same co-hosted BSSs that have enabled NPCA operation.

An NPCA AP that intends to enable, disable or update the parameters of NPCA mode shall follow the procedure defined in 37.32 (Enhanced BSS parameter critical update procedure(#12610))(#Ed) to notify its associated non-AP STAs.

An NPCA AP that has enabled NPCA operation shall set to 1, the NPCA Enabled field in the UHR (#12558)Operation element of the (Re)Association Response, UHR Link Reconfiguration Notify, Beacon and Probe Response frames that it transmits.

An NPCA AP with dot11HEPSROptionImplemented set to true shall set the TXVECTOR parameter SPATIAL_REUSE to PSR_DISALLOW for PPDU's that it transmits, and shall set the PSR Disallowed subfield in the SR Control field of the Spatial Reuse Parameter Set element to 1 in Management frames it transmits before enabling NPCA operation in its BSS and while NPCA operation remains enabled.

An AP enables PHYLEN NPCA operation by setting the MOPLN NPCA field to 0 and enables both PHYLEN NPCA and MOPLN NPCA operation by setting the MOPLN NPCA field to 1.

1 An NPCA AP may advertise an NPCA Disabled Subchannel Bitmap field in the NPCA Operation Parameters field that indicates the subchannels that are punctured when an NPCA STA operates on the NPCA primary channel:

- 2 — If an NPCA Disabled Subchannel Bitmap field is present, then the NPCA Disabled Subchannel Bitmap Field Present bit shall be set to 1, otherwise the NPCA Disabled Subchannel Bitmap Field Present field shall be set to 0.
- 3 — The NPCA Disabled Subchannel Bitmap field value shall satisfy the following requirements:
 - 4 • The puncturing pattern indicated by the value of the NPCA Disabled Subchannel Bitmap field is a valid non-OFDMA puncturing pattern as defined in 36.3.12.11.3 (Preamble puncturing for EHT MU PPDU in a non-OFDMA transmission)
 - 5 • A 20 MHz subchannel indicated as punctured in the Disabled Subchannel Bitmap field of the EHT Operation element (if any) is also indicated as punctured in the NPCA Disabled Subchannel Bitmap field.
- 6 — An NPCA AP may indicate one or more 20 MHz subchannels as punctured in the NPCA Disabled Subchannel Bitmap field of the EHT Operation Element for the purpose of maximizing the BW of the NPCA operating channel
- 7 — An NPCA AP may indicate one or more 20 MHz subchannels as punctured in the NPCA Disabled Subchannel Bitmap field of the EHT Operation Element for the purpose of creating a gap between the PPDU that initiated the NPCA switch and the NPCA operating channel
- 8 — If no NPCA Disabled Subchannel Bitmap field is present in the NPCA Operation Parameters field transmitted by the AP that the STA is associated with, then the subchannels are punctured during NPCA operation as indicated in 35.15.2 (Preamble puncturing operation).

9 An NPCA AP shall indicate a value in the NPCA Primary Channel field of transmitted NPCA Operation Parameters fields that corresponds to a channel that is located within the secondary 40 MHz of the BSS operating channel if the BSS is an 80 MHz BSS, that corresponds to a channel that is located within the secondary 80 MHz of the BSS operating channel if the BSS is a 160 MHz BSS and that corresponds to a channel that is located within the secondary 160 MHz of the BSS operating channel if the BSS is a 320 MHz BSS.

10 A non-AP NPCA STA shall indicate its NPCA switching delay and NPCA switch back delay respectively in the NPCA Switching Delay field and NPCA Switch Back Delay fields of the OMP Request frames.

11 When a non-AP STA that supports NPCA mode (re)associates with an AP, the NPCA mode is disabled by default for the non-AP STA. A UHR non-AP STA that supports NPCA mode and that intends to enable, disable or update the parameters of NPCA mode shall follow the procedure defined in 37.31 (Procedure for operating mode and parameter updates). In the UHR OMP request sent to enable or update the parameters of NPCA mode for the non-AP STA, the non-AP STA shall include the following in the Mode Parameters field of the Mode Tuple field:

- 12 — NPCA switching delay,
- 13 — NPCA switch back delay.

14 The associated AP shall accept the request and follow the procedure defined in 37.31 (Procedure for operating mode and parameter updates).

15 NOTE—For a non-AP STA to enable NPCA mode, the associated AP must support NPCA and must have NPCA enabled for the BSS (see 37.31 (Procedure for operating mode and parameter updates)).

16 37.18.2 MU EDCA interaction with NPCA

17 If an NPCA AP that has enabled NPCA operation advertises MU EDCA parameters in the Beacon frames that it transmits, the MU EDCA protocol (see 26.2.7 (EDCA operation using MU EDCA parameters)) shall apply jointly on both BSS primary channel and NPCA primary channel for a non-AP NPCA STA. An NPCA

STA shall follow the MU EDCA procedure in 26.2.7 (EDCA operation using MU EDCA parameters). In addition, an NPCA STA shall:

- Maintain a single MU EDCA timer that is shared across the BSS primary channel and the NPCA primary channel
- Transition from using EDCA parameters to using MU EDCA parameters (and vice-versa) at the same time on both the BSS primary channel and the NPCA primary channel based on conditions described in 26.2.7 (EDCA operation using MU EDCA parameters) that occur on either the BSS primary channel or the NPCA primary channel, and
- When the STA is operating on the NPCA primary channel, use the same MU EDCA parameters as are used on the BSS primary channel (dot11MUEDCATable) except that AIFSN[AC] shall be set to 0 for all ACs. When the STA switches back to the BSS primary channel, it reverts to using the AIFSN[AC] values from the dot11MUEDCATable.

37.18.3 Switching to the NPCA channel

An NPCA STA shall not switch to the NPCA primary channel for NPCA operation if NPCA mode has not been enabled by its associated AP according to the procedure described in 37.32 (Enhanced BSS parameter critical update procedure(#12610)).

An NPCA STA may switch to the NPCA primary channel for NPCA operation if the NPCA mode has been enabled for the BSS of which it is a member and either condition 1) or 2) is met:

- 1) the STA received a PPDU and/or received a PHY-RXSTART.indication primitive for an HE/EHT/UHR PPDU on the BSS primary channel and all of the following conditions are true:
 - a. Condition 2) is not true.
 - b. the PPDU is classified by the STA as an inter-BSS PPDU following the procedure defined in 37.5 (Intra-BSS and inter-BSS PPDU classification for UHR STA).
 - c. At least one of the following conditions is true:
 - i) The value of the MAC variable NPCA_PPDU_REM_DUR derived from the received PPDU is greater than the value indicated in the most recently received or transmitted NPCA Minimum Duration Threshold field corresponding to the BSS of which the STA is a member.
 - ii) If the NPCA AP corresponding to the BSS of which the STA is a member has enabled MOPLEN NPCA in addition to PHYLEN NPCA and the value of the MAC variable NPCA_PHY_TXOP_REM_DUR derived from the received PPDU is greater than the value indicated in the most recently received or transmitted NPCA Minimum Duration Threshold field corresponding to the BSS of which the STA is a member.
 - d. The bandwidth of the PPDU is determined by the STA to be 20, 40, 80, 160 or 320 MHz, based on the Bandwidth field in the PHY preamble of the PPDU, and the channel occupied by the PPDU does not overlap with the NPCA primary channel.
 - e. If the STA maintains an intra-BSS NAV, it is zero.
- 2) All of the following conditions are true:
 - a. A sequence of three PDUs, separated by aSIFSTime, is identified on the BSS primary channel, comprising an initial Control frame, an initial Response frame and a third PDU following the initial Response frame, where the presence of the initial Response frame may be inferred.
 - b. The STA received at least the first PDU containing the initial Control frame and the PHY-RXSTART.indication and/or the PHY-RXEARLYSIG.indication of the third PDU.

- c. An indication that a valid TXOP was obtained on the BSS primary channel, as verified by the receipt of a PHY-RXEARLYSIG.indication or PHYRXSTART.indication primitive corresponding to the third PPDU that occurs during a time window that:
 - i) begins at aSIFSTime + ICR_Timeout after the MAC receives a PHY-RXEND.indication primitive corresponding to the first PPDU, where ICR_Timeout is equal to:
 - 1) the length (in usec) of the expected CTS if the initial Control frame is an RTS or an MU-RTS Trigger frame,
 - 2) the value of RXTIME calculated using Equation (27-147) with the value of LENGTH replaced by the value from the UL Length field of the Common Info field, if the initial Control frame is a BSRP Trigger frame or a BSRP NTB Trigger frame
 - iii) has a duration that is equal to NPCA_START_TIMEOUT which is aSIFSTime + (2 x aSlotTime) + aRxPHYStartDelay
- d. At least one of the three PPDU in the sequence of PPDU is classified by the STA as an inter-BSS PPDU following the procedure defined in 37.5 (Intra-BSS and inter-BSS PPDU classification for UHR STA).
- e. At least one of the following conditions is true:
 - i) The NPCA AP corresponding to the BSS of which the STA is a member has enabled PHYLEN NPCA only and the value of the MAC variable NPCA_PPDU_REM_DUR derived from the received third PPDU of the sequence of PPDU is greater than the value indicated in the most recently received or transmitted NPCA Minimum Duration Threshold field corresponding to its BSS
 - ii) If the NPCA AP corresponding to the BSS of which the STA is a member has enabled MOPLen NPCA in addition to PHYLEN NPCA and the value of the MAC variable NPCA_CFRAME_TXOP_REM_DUR derived from the received first PPDU (containing the initial Control frame of the control frame exchange) of the sequence of PPDU is greater than the value indicated in the most recently received or transmitted NPCA Minimum Duration Threshold field corresponding to its associated BSS
- f. The bandwidth of the third PPDU is determined by the STA to be 20, 40, 80, 160 or 320 MHz based on the Bandwidth field in the PHY preamble of the PPDU not overlap with the NPCA primary channel and the channel occupied by the PPDU does not overlap with the NPCA primary channel.
- g. If the STA maintains an intra-BSS NAV, it is zero at the time of the receipt of the PHY-RXSTART.indication and/or the PHY-RXEARLYSIG.indication of the first PPDU.

When a PHY-CCA.indication(BUSY) primitive corresponding to the start of the reception of a PPDU is indicated at an NPCA STA while operating on the BSS primary channel, the values of the MAC variables NPCA_PPDU_REM_DUR, NPCA_PHY_TXOP_REM_DUR and NPCA_TIMER are all set to 0. When a PHY-CCA.indication(BUSY) corresponding to the start of the reception of a PPDU containing an initial Control frame is indicated at an NPCA STA while operating on the BSS primary channel, the MAC variable NPCA_CFRAME_TXOP_REM_DUR is set to 0.

The MAC variable NPCA_PPDU_REM_DUR derived from a received PPDU is equal to the value in microseconds, of the remaining duration of the received PPDU, determined by the MAC at the time of the receipt of the PHY-RXSTART.indication primitive associated with the received PPDU, by subtracting the time elapsed between the reception of the PHY-CCA.indication(BUSY) and PHY-RXSTART.indication primitives associated with the received PPDU from the value of RXTIME of the received PPDU.

The MAC variable NPCA_PHY_TXOP_REM_DUR derived from a received PPDU is:

- Set to 0, if the RXVECTOR parameter TXOP_DURATION is UNSPECIFIED, or if the NPCA AP corresponding to the BSS of which the STA is a member has not enabled MOPLen NPCA.

- Otherwise, it is set to the value in `usec`, of the remaining duration of the PPDU, determined by the MAC at the time of the receipt of the `PHY-RXSTART.indication` primitive associated with the received PPDU, by subtracting the time elapsed between the reception of the `PHY-CCA.indication(BUSY)` and `PHY-RXSTART.indication` primitives associated with the received PPDU from the value of `RXTIME` corresponding to the received PPDU, plus the value of the `TXOP_DURATION` parameter of the `RXVECTOR` of the PPDU.

The MAC variable `NPCA_CFRAME_TXOP_REM_DUR` derived from a received PPDU is:

- Set to 0, if the NPCA AP corresponding to the BSS of which the STA is a member has not enabled `MOPLen NPCA`
- Otherwise, it is set to the value in the `Duration/ID` field of the initial Control frame in the received PPDU at the receipt of the `PHY-RXEND.indication` primitive of the PPDU that contained the frame. The value of `NPCA_CFRAME_TXOP_REM_DUR` is reduced by the amount of time elapsed between the `PHY-RXEND.indication` primitive of the initial Control frame from which the value of `NPCA_CFRAME_TXOP_REM_DUR` was determined and the `PHY-RXSTART.indication` primitive of the third PPDU of the frame exchange sequence identified in condition 2) above at the time of the receipt of the `PHY-RXSTART.indication` primitive of the third PPDU.

37.18.4 NPCA transmission rules

When an NPCA STA switches to the NPCA primary channel for NPCA operation, then the following rules apply:

- 1) If the STA switches from the BSS primary channel to the NPCA primary channel based on meeting condition 1) of 37.18.3 (Switching to the NPCA channel), the STA shall initiate the switch at the NPCA HE switch time and it shall be ready to transmit and receive frames (subject to its capabilities and operating mode) on the NPCA primary channel no later than the value of its most recently indicated NPCA switching delay after the NPCA HE switch time. The NPCA HE switch time is the point in time immediately after the reception of the `HE-SIG-A/U-SIG` field of the received PPDU in condition 1) of 37.18.3 (Switching to the NPCA channel).
- 2) If the STA switches from the BSS primary channel to the NPCA primary channel based on meeting condition 2) of 37.18.3 (Switching to the NPCA channel), the STA shall initiate the switch at the NPCA NHT switch time and it shall be ready to transmit and receive frames addressed to it (subject to its capabilities and operating mode) on the NPCA primary channel no later than the value of its most recently indicated NPCA switching delay after the NPCA NHT switch time. The NPCA NHT switch time is equal to the point in time that is $3 \times \text{TSYM}$ (as defined in Table 17-5 - Timing related parameters, in the column labeled "Value (20 MHz channel spacing)") after the reception of the `L-SIG` field of the third PPDU of the received sequence of PPDUs from condition 2) of 37.18.3 (Switching to the NPCA channel).
- 3) The STA shall use the same EDCA parameter set and EPCS EDCA parameter set values for operation on the NPCA primary channel as it uses on the BSS primary channel. See 37.18.2 (MU EDCA interaction with NPCA) for MU EDCA parameters.
- 4) At each NPCA HE switch time or NPCA NHT switch time, as appropriate, if either the STA is a non-AP STA and transmission of frames that are not a response to a Trigger frame is not disabled by the MU EDCA protocol (See 26.2.7 (EDCA operation using MU EDCA parameters)) or the STA is an AP, the STA may initiate a TXOP on the NPCA primary channel by following the rules defined in 10.23.2.2 (EDCA backoff procedure) and 10.23.2.4 (Obtaining an EDCA TXOP) with the following exceptions:
 - a. Each time that the STA switches to the NPCA primary channel, the STA shall
 - i) If condition 1) from 37.18.3 (Switching to the NPCA channel) is met, set `NPCA_CFRAME_TXOP_REM_DUR` to 0, set `NPCA_TIMER` to the largest non-zero value of the variables `NPCA_PPDU_REM_DUR`, `NPCA_PHY_TXOP_REM_DUR` and

- 1 NPCA_CFRAME_TXOP_REM_DUR, minus the switch back delay that the STA
2 has indicated in the most recently transmitted NPCA Operation Parameters field.
3 ii) Store the current values of the variables QSRC[AC], CW[AC] and the backoff
4 counter for each EDCAF.
5 iii) Set QSRC[AC] for each AC to the value of the Initial NPCA QSRC field of the
6 NPCA Operation Parameters received from its associated NPCA AP or that it trans-
7 mitted.
8 iv) Initialize variables CW[AC] to $2^{\text{Init_QSRC_NPCA}} \times (\text{CWmin}[AC] + 1) - 1$.
9 v) Invoke the backoff procedure even if the medium for the NPCA primary channel is
10 not busy.
11 vi) Initiate countdown of the MAC variable NPCA_TIMER in units of 1 μs
12
13 5) A first STA shall not initiate a transmission on the NPCA primary channel to a second STA
14 until the second STA's NPCA switching delay time has elapsed since the NPCA HE switch
15 time at the first STA if the first STA is switching due to condition 1) above or since the NPCA
16 NHT switch time at the first STA if the first STA is switching due to condition 2) above.
17
18 6) The STA shall begin all frame exchanges on the NPCA primary channel with an ICF using
19 non-HT PPDU or non-HT duplicate PPDU format using a rate of 6 Mb/s, 12 Mb/s, or 24 Mb/s.
20
21 a. For TXOPs initiated by an AP, the initial Control frame (ICF) shall be a BSRP Trigger
22 frame or an MU-RTS Trigger frame except when at least one of the target non-AP STA(s)
23 is operating in the DUO mode, in which case, the ICF may be a BSRP Trigger frame or a
24 BSRP NTB Trigger frame but not an MU-RTS. In addition:
25 i) The ICF shall conform to the rules in 37.19.2 (Dynamic (#7566)unavailability operation
26 (DUO) mode) if at least one of the target non-AP STA(s) is operating in DUO
27 mode.
28 ii) The ICF shall conform to the rules in 37.21 (Enhanced multi-link single-radio
29 (EMLSR) operation for a UHR non-AP MLD) if at least one of the target non-AP
30 STA(s) is affiliated with a non-AP MLD that is operating in EMLSR mode.
31 iii) The ICF shall conform to the rules in 37.17.1 (Dynamic power save (DPS) operation)
32 if at least one of the target non-AP STA(s) is operating in DPS mode.
33
34 b. For TXOPs initiated by a non-AP STA, the initial control frame shall be a BSRP NTB
35 Trigger frame, except that if the non-AP STA is operating in the Dynamic Unavailability
36 Operation mode (DUO), then the ICF shall conform to the rules found in 37.19.2
37 (Dynamic (#7566)unavailability operation (DUO) mode).
38
39 7) An NPCA AP that transmits a Trigger frame on the NPCA primary channel shall indicate RU
40 index values that use the NPCA primary channel as the reference primary channel.
41
42 8) An NPCA STA that transmits a Trigger frame on the NPCA primary channel shall set the
43 NPCA Primary Indication field to 1 in the Special User Info field, otherwise, this field is set to
44 0.
45
46 9) The 20 MHz channels occupied by PPDU's transmitted by the STA shall meet all of the
47 following conditions:
48
49 a. include at least the NPCA primary channel.
50
51 b. all be within the BSS bandwidth.
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53 c. not include any of the channels occupied by either the PPDU of condition 1) of 37.18.3
54 (Switching to the NPCA channel) or by the third PPDU of condition 2) of 37.18.3
55 (Switching to the NPCA channel), whichever caused the STA to switch from the BSS
56 primary channel to the NPCA primary channel.
57
58 d. not include channels that are indicated as punctured in the Disabled Subchannel Bitmap
59 field in the EHT Operation element or in the NPCA Disabled Subchannel Bitmap field in
60 the UHR Operation element with the following rules:
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62
63
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- i) If the associated NPCA AP advertises an NPCA Disabled Subchannel Bitmap field, the STA shall follow the rules in 35.15.2 (Preamble puncturing operation) except that instead of the Disabled Subchannel Bitmap field it shall use the most recently exchanged NPCA Disabled Subchannel Bitmap field.
 - ii) If the associated NPCA AP does not transmit an NPCA Disabled Subchannel Bitmap field, the STA shall follow the rules in 35.15.2 (Preamble puncturing operation).
- 10) UHR ELR PPDUs, HE ER SU PPDUs, EHT MCS14/15 shall not be transmitted on the NPCA primary channel.
 - 11) Dynamic Subband Operation (see 37.26 (Dynamic subband operation (DSO)(#7567))) shall not be used on the NPCA primary channel.
 - 12) If TBTT for the BSS occurs while an NPCA AP is operating on the NPCA primary channel, the scheduling of the transmission of the Beacon frame and following group addressed frames shall be deferred until immediately after the AP switches back to the BSS primary channel.

NOTE—The AP and associated STAs are not required to switch back to the BSS primary channel at TBTT. The group addressed frames will be buffered and delivered immediately following the next DTIM Beacon, unless explicitly specified otherwise.

NOTE—Exponential backoff applies on the NPCA primary channel when there are failed transmissions.

37.18.5 Switching back from the NPCA channel

An NPCA STA shall switch back to the BSS primary channel when the NPCA_TIMER expires.

When the STA switches back to the BSS primary channel, it shall:

- 1) replace the current values of the variables QSRC[AC], CW[AC] and the backoff counter for each EDCAF with the values that it stored when it switched to the NPCA primary channel.
- 2) resume the backoff procedure.

37.19 Unavailability reporting and parameter updates

37.19.1 General

The unavailability reporting and parameter updates subclause describes a set of mechanisms that allow a STA to inform a peer STA of its unavailabilities and/or its adaptive operation parameters. Subclause 37.19.2 (Dynamic (#7566)unavailability operation (DUO) mode) describes how a non-AP STA can indicate unavailability using certain Control frames, where the unavailability duration might overlap with the ongoing TXOP. Subclause 37.19.3 (Non-AP STA periodic unavailability operation (PUO) mode) describes how a non-AP STA can inform its associated AP about periodic service periods where the non-AP STA will be unavailable. Subclause 37.19.4 (AP PUO mode) describes how and under which conditions an AP can operate its BSS with periodic service period(s) during which the AP will be unavailable. Finally, Subclause 37.19.5 (Adaptive operation mode) describes a mode of operation that allows a non-AP STA to adapt its operation parameters when experiencing in-device coexistence issues.

37.19.2 Dynamic (#7566)unavailability operation (DUO) mode

A non-AP STA that has dot11DUOOptionImplemented equal to true supports DUO, is called a DUO non-AP STA and shall set the DUO Support field of the UHR MAC Capabilities Information field of the UHR Capabilities element to 1. An AP that has dot11DUOAssistingOptionImplemented equal to true supports DUO, is called a DUO assisting AP and shall set the DUO Support field of the UHR MAC Capabilities Information field of the UHR Capabilities element to 1.

When a DUO non-AP STA (re)associates with an AP, DUO mode is disabled by default for the DUO non-AP STA. A DUO non-AP STA that intends to enable or disable DUO mode shall follow the procedure